

Ex.no:8	MINI PROJECT – FINANCIAL DATA ANALYTICS

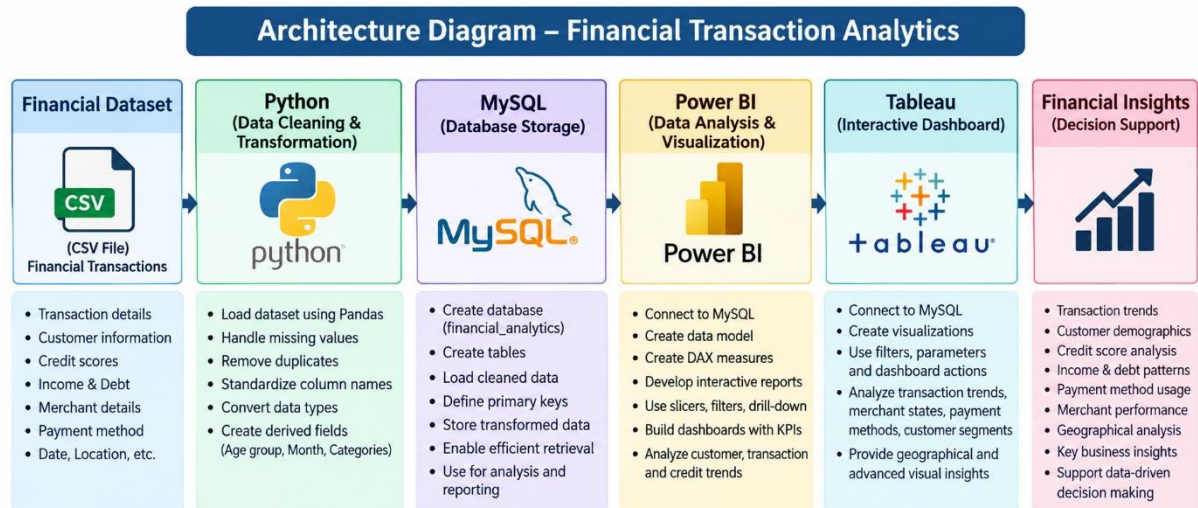
Aim:

To clean, transform, store, and analyze financial transaction data using Python, MySQL, Power BI, and Tableau to derive meaningful insights into transaction trends, customer demographics, credit scores, income levels, debt patterns, payment methods, and merchant performance.

Algorithm:

1. **Load** the financial transaction dataset from the CSV file using Pandas.
2. **Clean** the data by handling missing values, removing duplicate records, standardizing column names, and correcting data types.
3. **Transform** the data by preparing transaction dates and deriving useful fields such as transaction year, transaction month, age groups, credit categories, debt categories, and income categories.
4. **Store** the cleaned and transformed dataset in a MySQL database named financial_analytics.
5. **Connect** the cleaned financial dataset to Power BI and Tableau for data analysis and visualization.
6. **Create** DAX measures and calculated fields for key metrics such as total transactions, total transaction value, average transaction value, average credit score, average yearly income, and total debt.
7. **Develop** interactive visualizations to analyze transaction trends, credit categories, customer demographics, debt patterns, payment methods, merchant states, and financial performance.
8. **Implement** KPIs, cards, slicers, filters, drill-down, drill-through, parameters, and interactive dashboard features in Power BI and Tableau.
9. **Generate** meaningful financial and customer insights from the Power BI and Tableau dashboards.
10. **End** the process.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

from sqlalchemy import create_engine, text

# Load dataset

df = pd.read_csv("Customer_financial_profiles.csv")

# Create a copy for processing

financial = df.copy()

# Remove duplicate records

financial.drop_duplicates(inplace=True)
```

```
# Handle missing values
```

```
financial["gender"] = financial["gender"].fillna("Unknown")
```

```
financial["address"] = financial["address"].fillna("Unknown")
```

```
financial["merchant_city"] = financial["merchant_city"].fillna("Unknown")
```

```
financial["merchant_state"] = financial["merchant_state"].fillna("Unknown")
```

```
financial["use_chip"] = financial["use_chip"].fillna("Unknown")
```

```
# Fill numerical missing values
```

```
numeric_columns = [
```

```
    "current_age",
```

```
    "birth_year",
```

```
    "birth_month",
```

```
    "per_capita_income",
```

```
    "yearly_income",
```

```
    "total_debt",
```

```
    "credit_score",
```

```
    "num_credit_cards",
```

```
    "amount"
```

```
]
```

```
for col in numeric_columns:
```

```
    if col in financial.columns:
```

```
        financial[col] = financial[col].fillna(financial[col].median())
```

```
# Convert date column
```

```
financial["date"] = pd.to_datetime(
```

```
    financial["date"],
```

```
    errors="coerce"
```

```
)
```

```
# Create transaction year
```

```
financial["transaction_year"] = financial["date"].dt.year
```

```
# Create transaction month
```

```
financial["transaction_month"] = financial["date"].dt.month
```

```
# Create age group
```

```
financial["age_group"] = pd.cut(  
    financial["current_age"],  
    bins=[17, 25, 35, 45, 55, 100],  
    labels=["18-25", "26-35", "36-45", "46-55", "56+"]  
)
```

```
# Standardize column names
```

```
financial.columns = (  
    financial.columns  
    .str.lower()  
    .str.strip()  
    .str.replace(" ", "_")  
)
```

```
# Display dataset information
```

```
print("Original Shape:", df.shape)
```

```
print("Cleaned Shape:", financial.shape)
```

```
print(financial.head())
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

MySQL Command Prompt

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL

```
CREATE DATABASE IF NOT EXISTS financial_analytics;
```

```
USE financial_analytics;
```

Python/Jupyter

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(
```

```
    f'mysql+pymysql://{username}:{password}@{host}:{port}/financial_analytics'
```

```
)
```

```
# Load cleaned data into MySQL
```

```
financial.to_sql(
```

```
    "financial_transactions",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```
    index=False,
```

```
    chunksize=5000
```

```
)
```

```
print("Data loaded successfully into MySQL.")
```

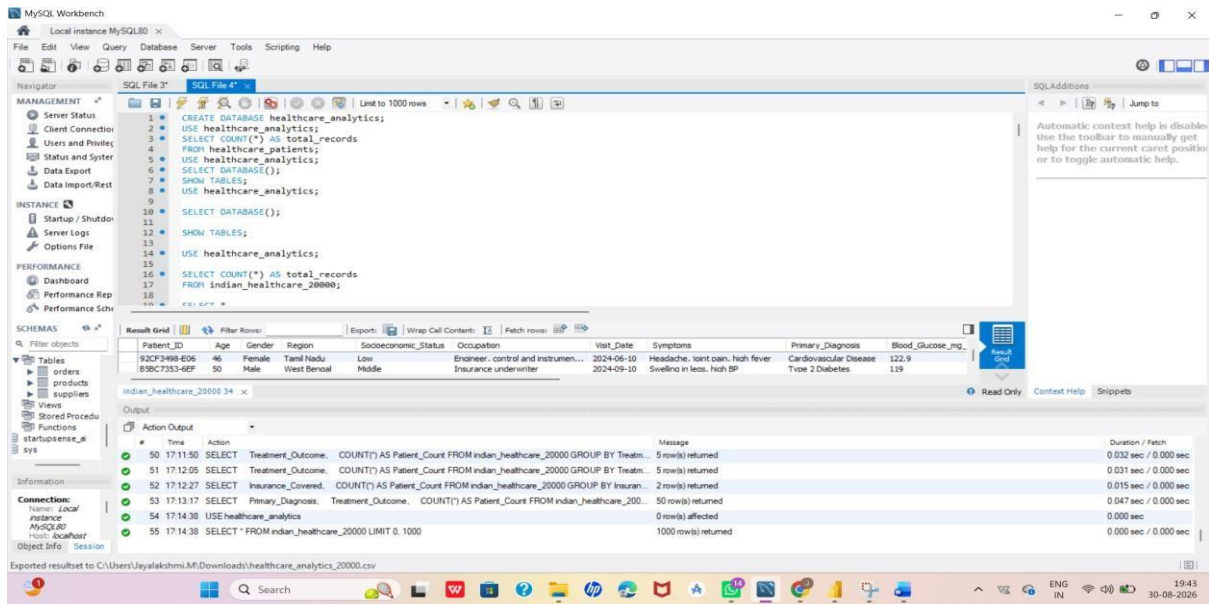
Verify

```
pd.read_sql(
```

```
    "SELECT COUNT(*) AS total_rows FROM financial_transactions",
```

```
    db_engine
```

```
)
```



3. Power BI – DAX Measures

Total Transactions =

COUNTROWS(FinancialData)

Total Transaction Value =

SUM(FinancialData[amount])

Average Transaction =

AVERAGE(FinancialData[amount])

Average Credit Score =

AVERAGE(FinancialData[credit_score])

Average Yearly Income =

AVERAGE(FinancialData[yearly_income])

Total Debt =

SUM(FinancialData[total_debt])

Average Credit Cards =

AVERAGE(FinancialData[num_credit_cards])

4. Power Bi Dashboard

Created visuals in Power Bi in separate sheets and created a dashboard

Output:

Original rows: 20000
Cleaned rows: 20000
Columns: 28

Missing values:
id 0 current_age 0 birth_year 0 birth_month 0 gender 0 address 0
per_capita_income 0 yearly_income 0 total_debt 0 credit_score 0 credit_category 0 debt_category 0
income_category 0 num_credit_cards 0 transaction_id 0 date 0 client_id 0 card_id 0
amount 0 use_chip 0 merchant_id 0 merchant_city 0 merchant_state 0 zip 0
transaction_year 0 transaction_month 0 age_group 0
dtype: int64

First 5 records:

	id	current_age	birth_year	birth_month	gender	address	per_capita_income	yearly_income	total_debt	credit_score	amount	use_chip	merchant_state	transaction_year
0	1	25	1999	7	F	123 Main St	45000	85000	15000	720	120.50	Swipe	CA	2023
1	2	34	1990	3	M	456 Oak Ave	52000	105000	32000	680	35.20	Chip	TX	2023
2	3	28	1996	11	F	789 Pine Rd	61000	95000	18000	745	89.99	Online	NY	2023
3	4	41	1983	5	M	321 Maple St	40000	78000	25000	690	15.00	Chip	FL	2023
4	5	37	1987	9	F	654 Cedar Blvd	58000	110000	40000	770	260.75	Swipe	IL	2023

[10]:
pd.read_sql(
"SELECT COUNT(*) AS total_rows FROM ipl_matches",
db_engine
)

[10]:

	total_rows
0	278205

[11]:
pd.read_sql(
"SELECT * FROM ipl_matches LIMIT 5",
db_engine
)

[11]:

	match_id	season	date	city	venue	team1	team2	toss_winner	winner	player_of_match	...	runs_scored	wickets_down	current_run_rate	required_run_rate	wicket_kind
0	335982	2008	2008-04-18	Bangalore	M. Chinnaswamy Stadium	RCB	KXIP	RCB	RCB	R. Dravid	...	4	0	12.00	10.50	NaN
1	335982	2008	2008-04-18	Bangalore	M. Chinnaswamy Stadium	RCB	KXIP	RCB	RCB	R. Dravid	...	1	0	10.50	11.00	NaN
2	335982	2008	2008-04-18	Bangalore	M. Chinnaswamy Stadium	RCB	KXIP	RCB	RCB	R. Dravid	...	0	0	9.50	11.20	NaN
3	335982	2008	2008-04-18	Bangalore	M. Chinnaswamy Stadium	RCB	KXIP	RCB	RCB	R. Dravid	...	6	0	10.00	11.00	NaN
4	335982	2008	2008-04-18	Bangalore	M. Chinnaswamy Stadium	RCB	KXIP	RCB	RCB	R. Dravid	...	0	1	9.00	10.80	bowled

5 rows x 34 columns

Navigator

Display Options ▾

localhost:3306 [1]

financial_analytics [1]

☒ financial_transactions

financial_transactions

id	current_age	birth_year	birth_month	gender	address	per_capita_income	yearly_income	total_debt	credit_score	...
1	25	1999	7	F	123 Main St	45000	85000	15000	720	...
2	34	1990	3	M	456 Oak Ave	52000	105000	32000	680	...
3	28	1996	11	F	789 Pine Rd	61000	95000	18000	745	...
4	41	1983	5	M	321 Maple St	40000	78000	25000	690	...
5	37	1987	9	F	654 Cedar Blvd	58000	110000	40000	770	...
6	52	1972	1	M	987 Elm St	62000	125000	60000	710	...
7	23	2001	6	F	147 Lake View	38000	62000	12000	650	...
8	45	1979	10	M	258 Hill Rd	50000	98000	35000	700	...
9	31	1993	4	F	369 Park Ave	47000	88000	21000	730	...
10	29	1995	8	M	741 River St	55000	102000	28000	690	...

The data in the preview has been truncated due to size limits.

Select Related Tables

Load

Transform Data

Cancel

File Home Help Table tools

Share ▾

Name FinancialData

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

Auto recovery contains some recovered files that haven't been opened.

View recovered files

Data

id	current_age	birth_year	birth_month	gender	address	per_capita_income	yearly_income	total_debt	credit_score
153	56	1969	1	Male	Street 257, Indore, Madhya Pradesh - 452346	90444	1085328	0	
243	56	1969	2	Male	Street 68, Kolkata, West Bengal - 700995	33325	399901	0	
307	57	1968	1	Male	Street 190, Ahmedabad, Gujarat - 380114	60775	729303	0	
586	65	1960	12	Male	Street 289, Nagpur, Maharashtra - 440407	37600	451198	0	
659	63	1962	3	Male	Street 170, Pune, Maharashtra - 411486	55493	665915	0	
857	65	1960	4	Male	Street 272, Kolkata, West Bengal - 700710	58901	706809	0	
990	65	1960	4	Male	Street 27, Pune, Maharashtra - 411487	52533	630393	0	
1482	64	1961	2	Male	Street 243, Nagpur, Maharashtra - 440245	37526	450312	0	
1655	57	1968	1	Male	Street 190, Ahmedabad, Gujarat - 380114	60775	729303	0	
1944	64	1961	2	Male	Street 243, Nagpur, Maharashtra - 440245	37526	450312	0	
1971	56	1969	1	Male	Street 257, Indore, Madhya Pradesh - 452346	90444	1085328	0	
1992	63	1962	4	Male	Street 73, Pune, Maharashtra - 411918	85472	1025670	0	
2121	59	1966	6	Male	Street 196, Surat, Gujarat - 395495	66959	803513	0	
2145	65	1960	4	Male	Street 272, Kolkata, West Bengal - 700710	58901	706809	0	
2371	56	1969	1	Male	Street 257, Indore, Madhya Pradesh - 452346	90444	1085328	0	
2421	60	1965	5	Male	Street 32, Bengaluru, Karnataka - 560811	38137	457641	0	
2501	59	1966	6	Male	Street 196, Surat, Gujarat - 395495	66959	803513	0	
2562	63	1962	4	Male	Street 73, Pune, Maharashtra - 411918	85472	1025670	0	
2585	63	1962	2	Male	Street 246, Mumbai, Maharashtra - 400463	46784	561403	0	
2613	65	1960	12	Male	Street 289, Nagpur, Maharashtra - 440407	37600	451198	0	
2641	56	1969	10	Male	Street 57, Bengaluru, Karnataka - 560943	21920	263046	0	
2671	64	1961	2	Male	Street 243, Nagpur, Maharashtra - 440245	37526	450312	0	
2675	63	1962	12	Male	Street 181, Hyderabad, Telangana - 500659	115532	1386383	0	
2949	57	1968	1	Male	Street 190, Ahmedabad, Gujarat - 380114	60775	729303	0	
3098	63	1962	2	Male	Street 246, Mumbai, Maharashtra - 400463	46784	561403	0	

Table: FinancialData (20,000 rows)

Data Visualization Using Power BI

The financial dataset was imported into Power BI Desktop to create interactive visualizations for analyzing transaction trends, customer demographics, credit scores, debt patterns, and merchant performance.

Visualization 1 — KPI Cards: Card

Displays Total Transactions, Total Transaction Value, Average Transaction, and Average Credit Score, providing a quick overview of key financial metrics.

Visualization 2 — Monthly Transaction Trend: Line Chart

Shows monthly changes in transaction amounts to identify financial trends over time.

Visualization 3 — Transaction Value by Credit Category: Stacked Column Chart

Compares transaction values across credit categories and genders.

Visualization 4 — Transaction Value by State: Horizontal Bar Chart

Displays transaction values across merchant states to identify high-performing regions.

Visualization 5 — Transaction Distribution by Payment Method: Donut Chart

Shows the proportion of transactions made through different payment methods.

Visualization 6 — Transaction Value by Age Group: Horizontal Bar Chart

Compares transaction values across different customer age groups.

Visualization 7 — Total Debt by Age Group: Area Chart

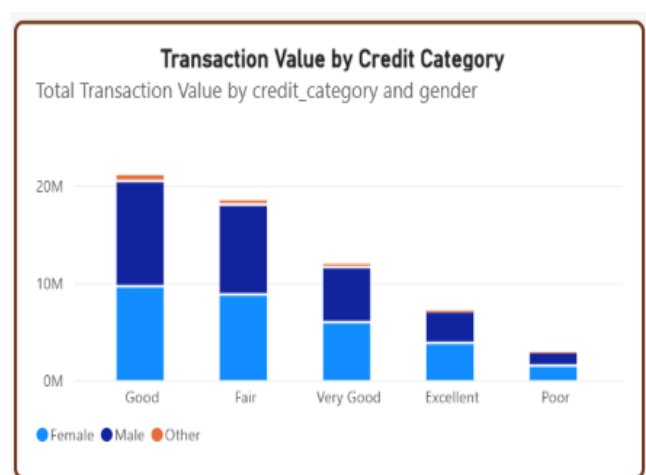
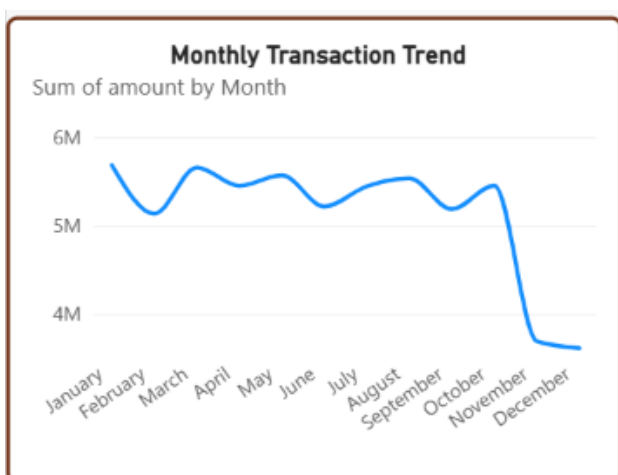
Represents debt distribution among different age groups.

Visualization 8 — Transaction Value by Year and Credit Category: Ribbon Chart

Shows changes in transaction values across years and credit categories.

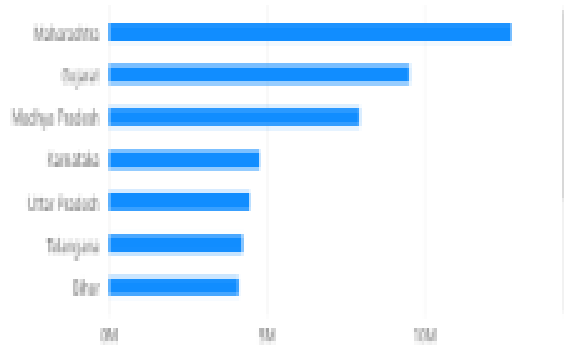
Visualization 9 — Transaction Distribution by Debt Category: Pie Chart

Displays the distribution of customers across different debt categories.



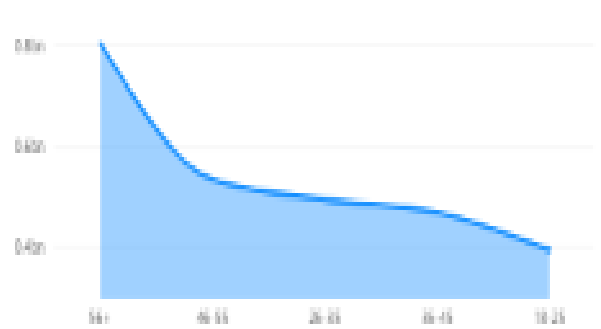
Transaction Value by State

Total Transaction Value by merchant_state



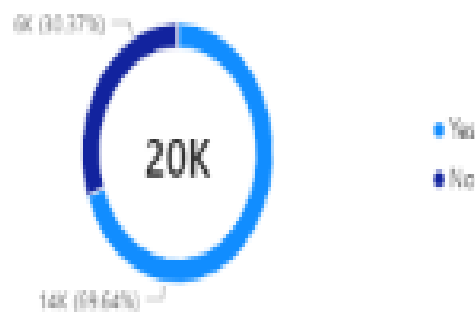
Total Debt by Age Group

Sum of total_debt by age_group



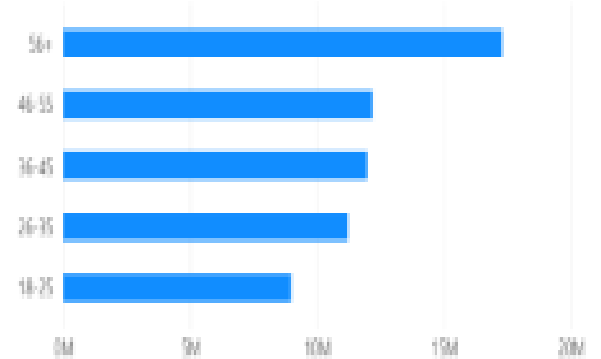
Transaction Distribution by Payment Method

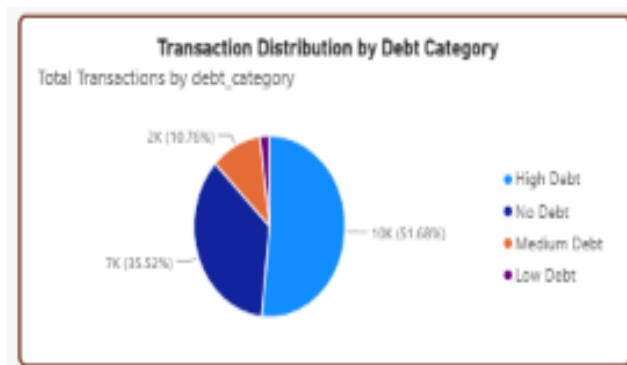
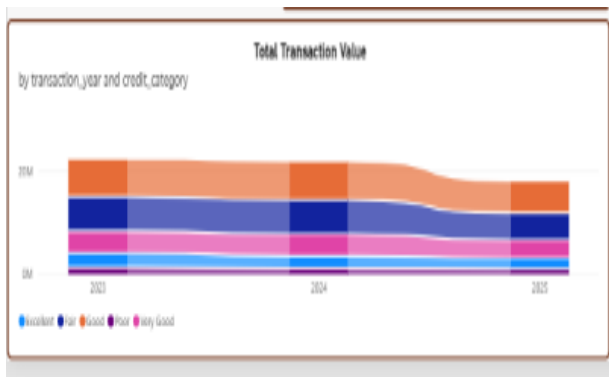
Total Transactions by use_chip



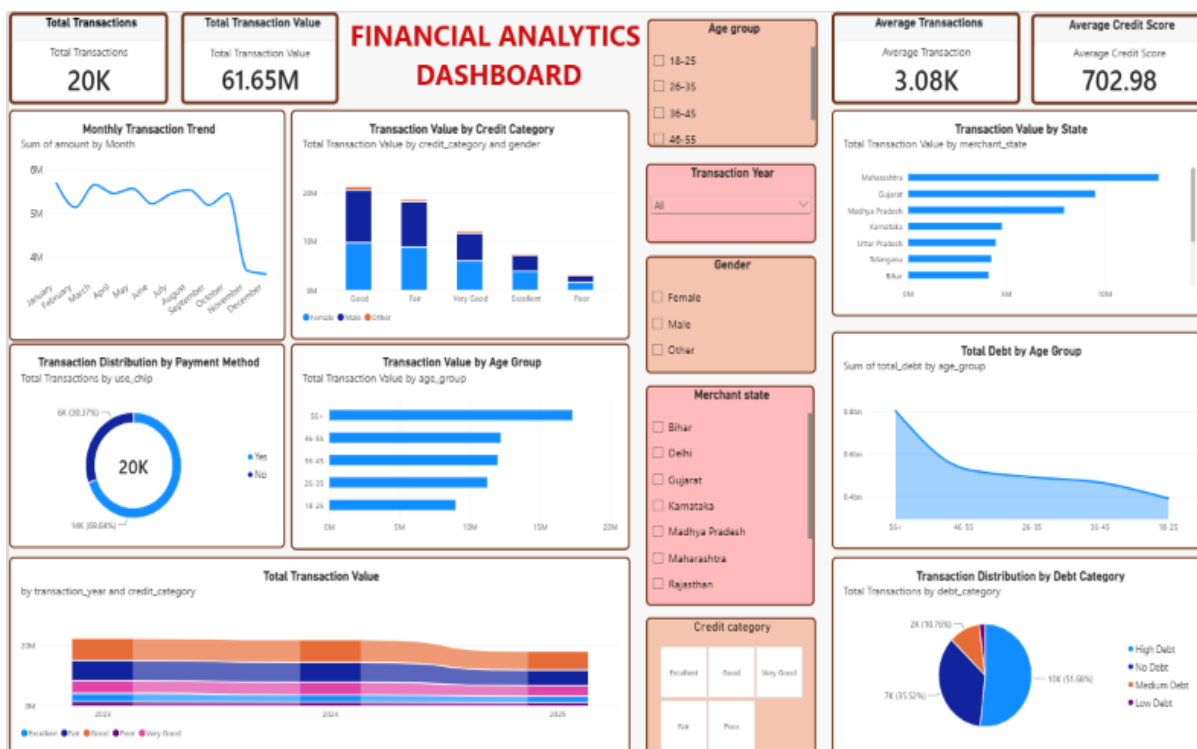
Transaction Value by Age Group

Total Transaction Value by age_group





FINANCIAL ANALYSIS POWER BI DASHBOARD

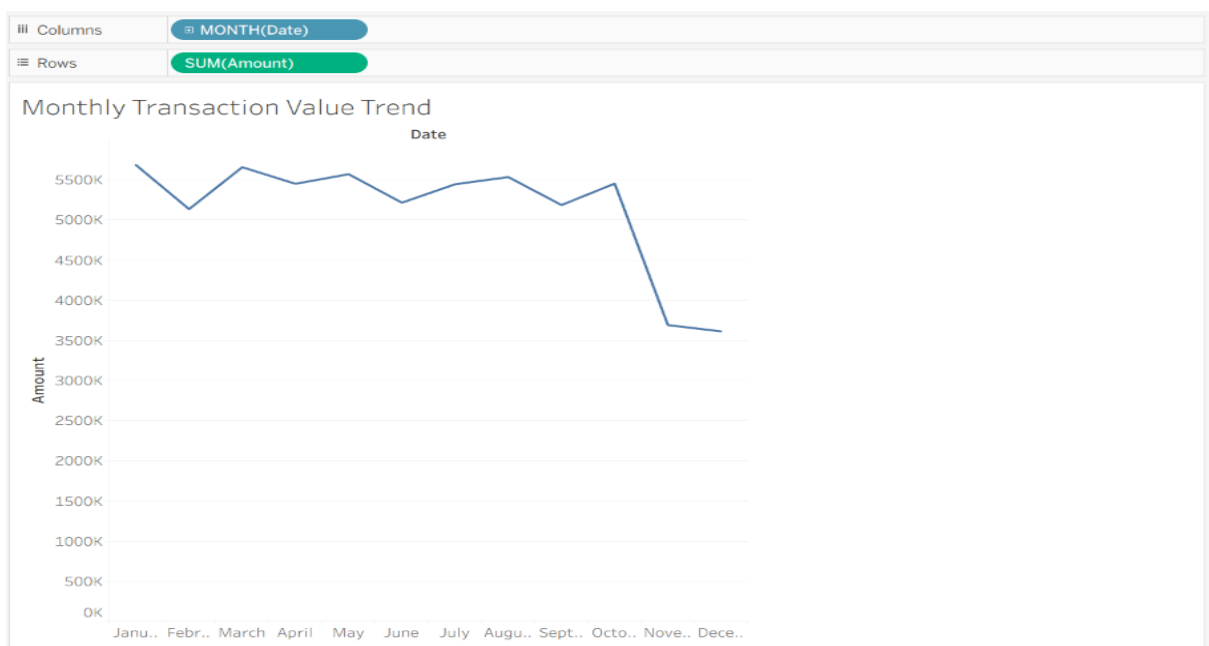


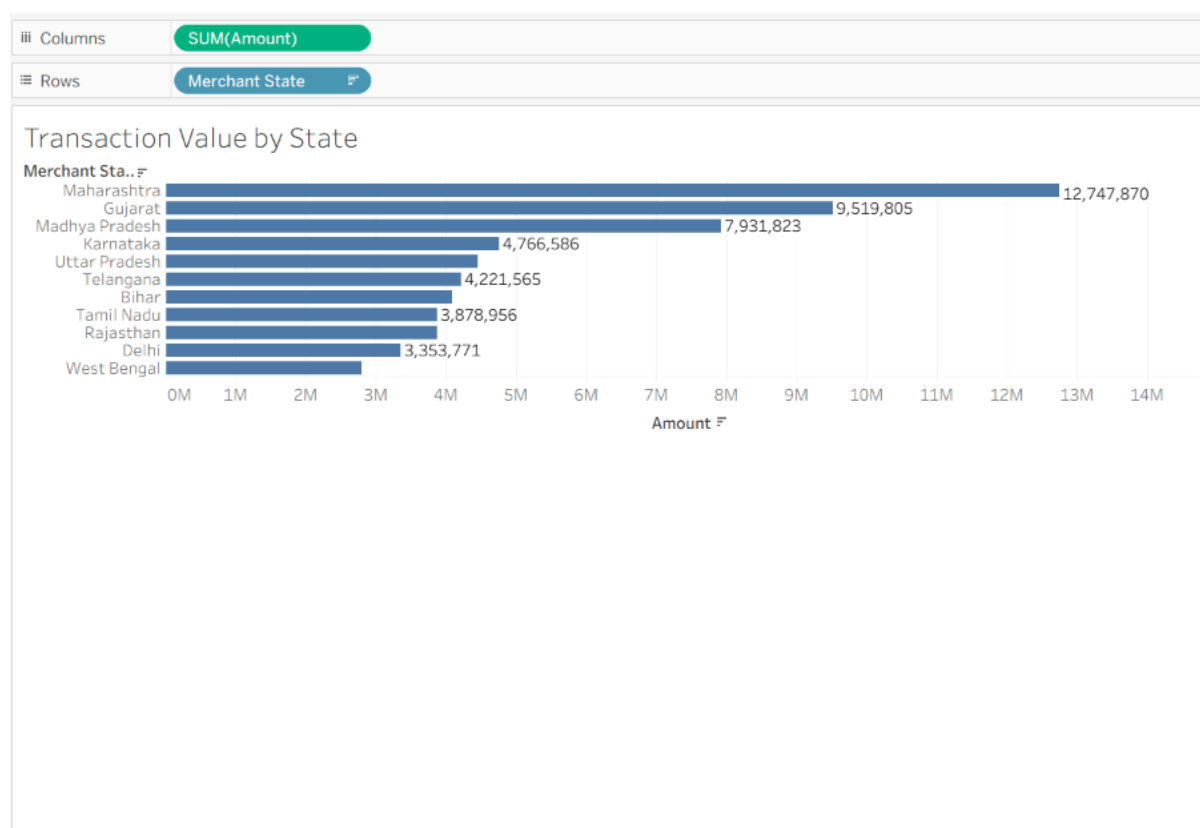
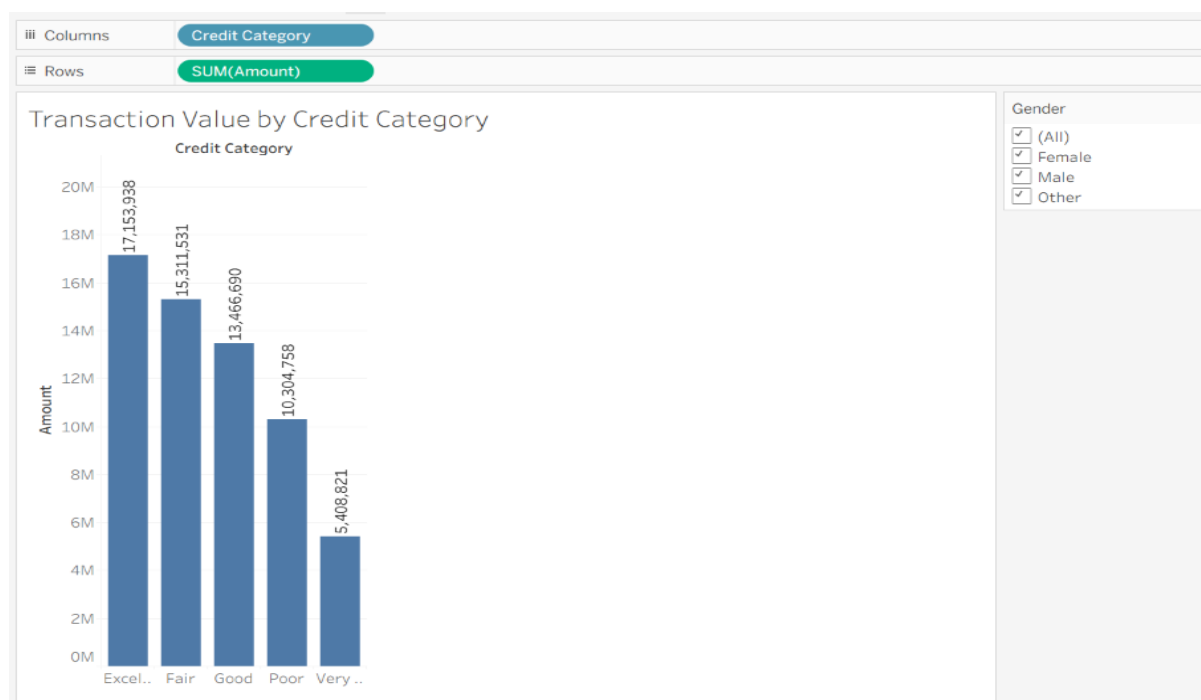
5. Tableau Dashboard

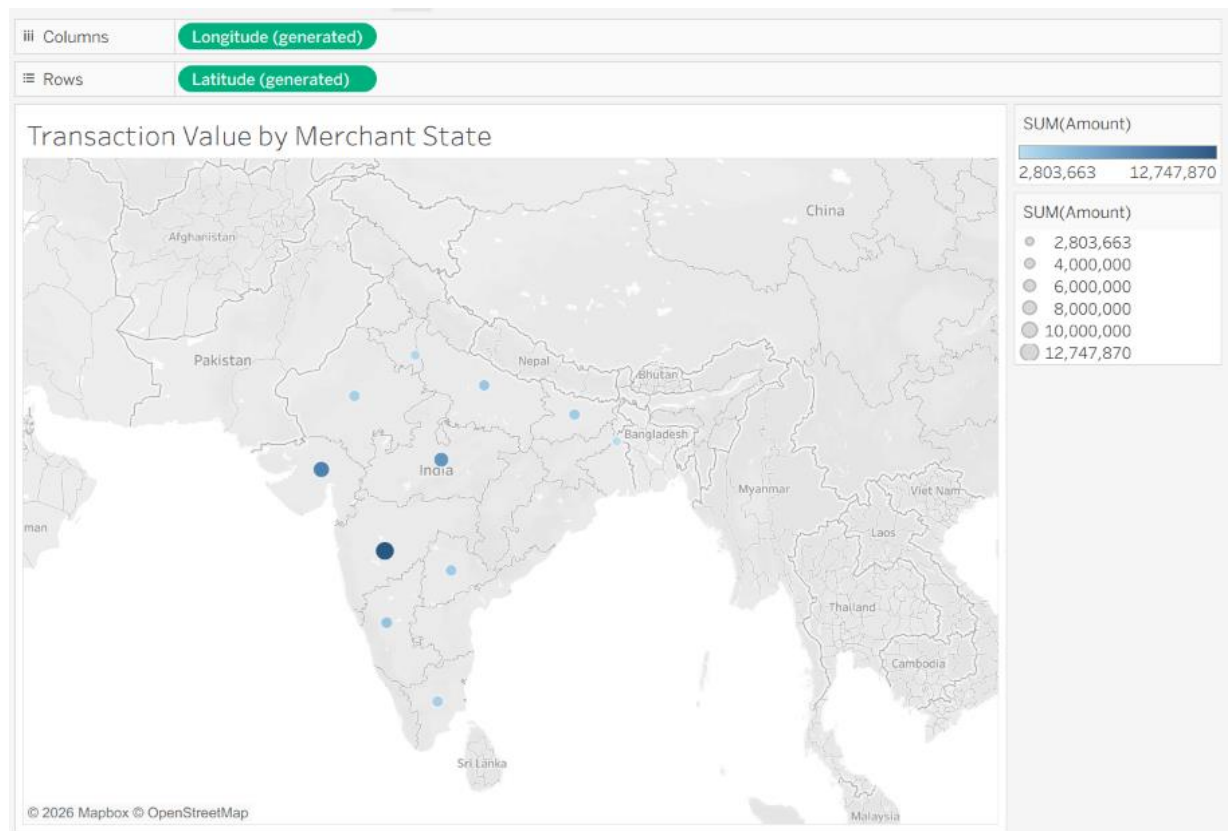
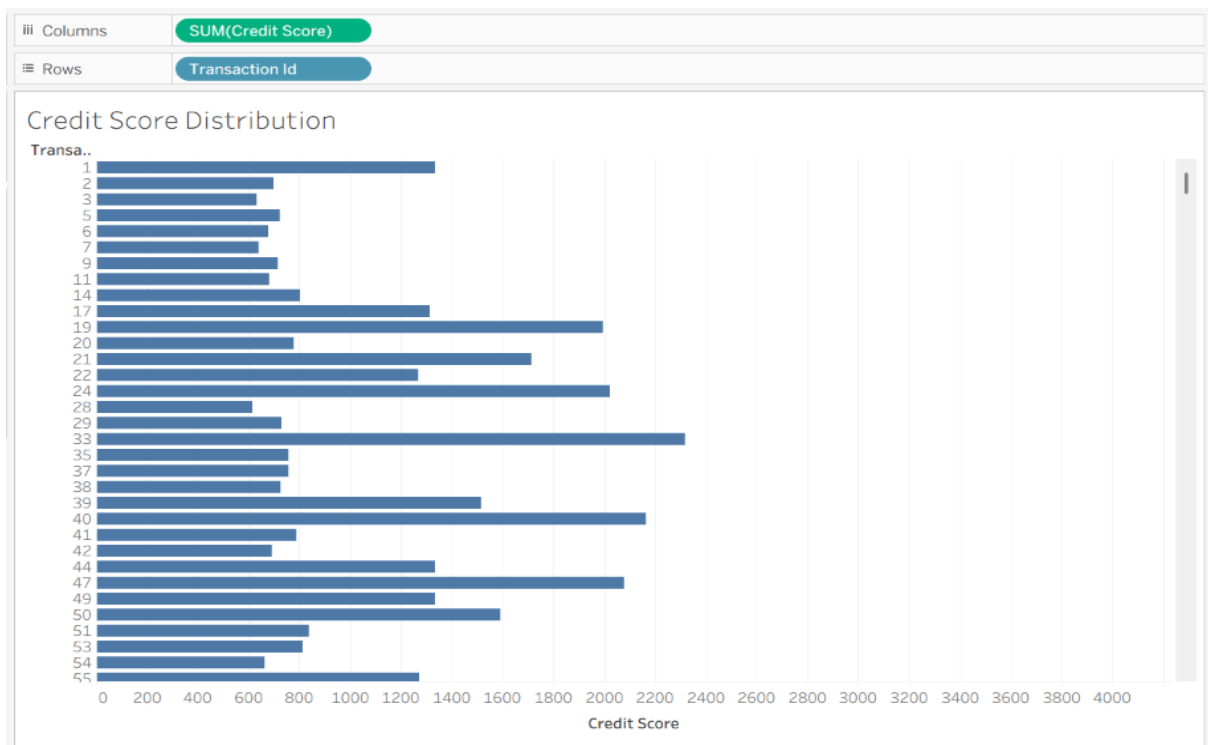
Created visuals in Tableau in separate sheets and created a dashboard

The screenshot displays the Tableau 'financial_transactions' table interface. On the left, the 'Connections' pane shows 'financial_analytics MySQL' and the 'Database' pane shows 'financial_analytics'. The 'Table' pane lists 'financial_transactions' with a 'New Custom SQL' option. The main area shows the table structure with 28 fields and 20,000 rows. A preview of the first 10 rows is shown below.

#	financial_transactions: id	financial_transactions: current_age	financial_transactions: birth_year	financial_transactions: birth_month	financial_transactions: gender	financial_transactions: address	financial_transactions: per_capita_income	financial_transactions: yearly_income	financial_transactions: total_debt
1	25	1999	7	F	123 Main St	45000	85000	15000	
2	34	1990	3	M	456 Oak Ave	52000	105000	32000	
3	28	1996	11	F	789 Pine Rd	61000	95000	18000	
4	41	1983	5	M	321 Maple St	40000	78000	25000	
5	37	1987	9	F	654 Cedar Blvd	58000	110000	40000	
6	52	1972	1	M	987 Elm St	62000	125000	60000	
7	23	2001	6	F	147 Lake View	38000	62000	12000	
8	45	1979	10	M	258 Hill Rd	50000	98000	35000	
9	31	1993	4	F	369 Park Ave	47000	88000	21000	
10	29	1995	8	M	741 River St	55000	102000	28000	



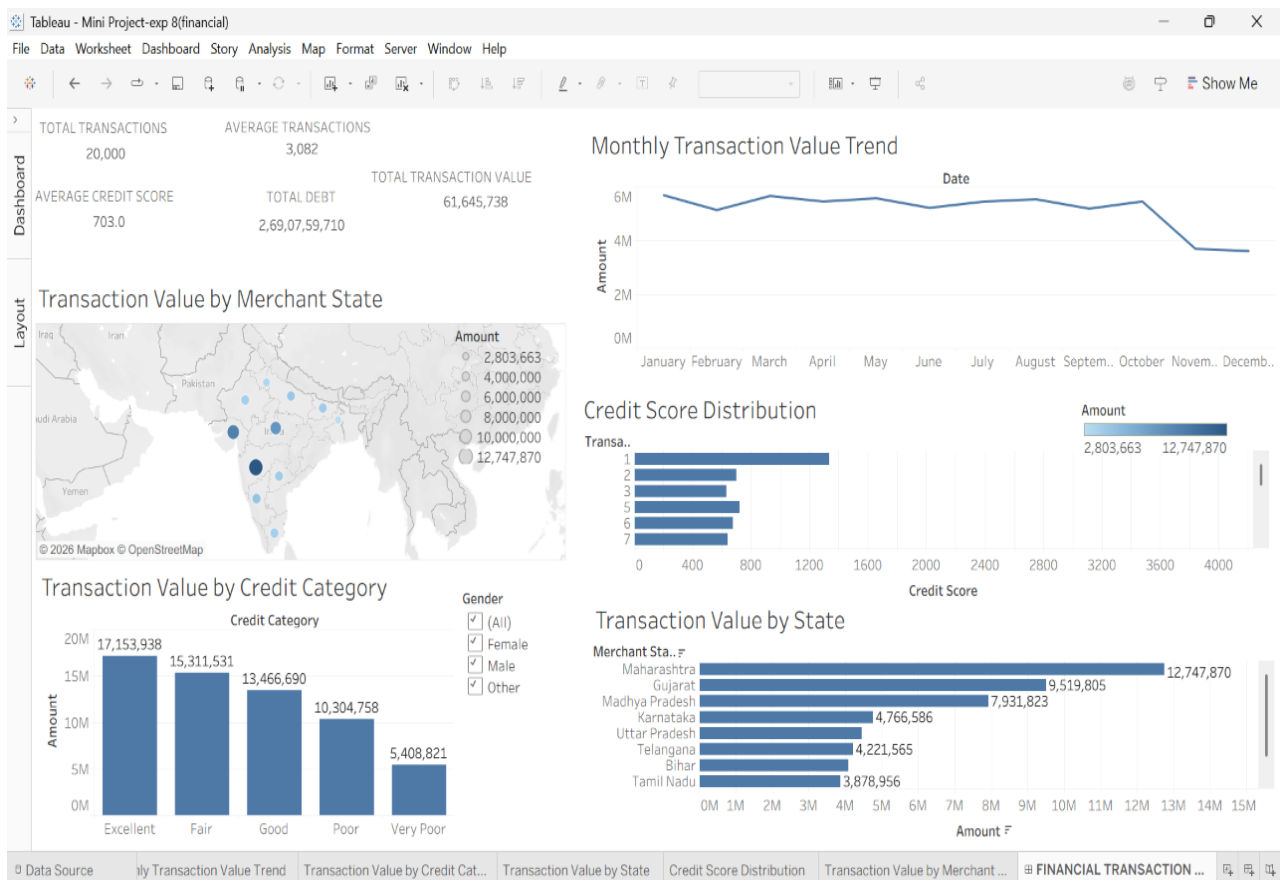




KPI

TOTAL TRANSACTIONS	20,000	AVERAGE TRANSACTIONS	3,082	
AVERAGE CREDIT SCORE	703.0	TOTAL DEBT	2,69,07,59,710	TOTAL TRANSACTION VALUE
				61,645,738

FINANCIAL ANALYSIS TABLEAU DASHBOARD



Result: